

AI CRM Lead Automation

A safe, testable workflow that converts inbound messages into structured CRM records, priorities, follow-up dates, and spreadsheet-safe exports.

DELIVERY	ANALYSIS	QUALITY	DEPLOYMENT
Local dashboard + API	Offline or optional OpenAI	69 automated tests	Local-first application

The business problem

Small teams receive inquiries through forms, inboxes, chat, and referrals. Manual interpretation and copy-paste work delay responses, lose context, and make it hard to see which lead needs attention first.

The implemented solution

The local application validates each intake, extracts contact details, classifies the request, calculates a transparent Hot/Warm/Cold score, schedules follow-up, checks duplicate emails, and writes an audit event. A responsive dashboard, authenticated JSON endpoints, lifecycle controls, and CSV export make the result immediately reviewable.

Two analysis modes, one contract

DEFAULT: DETERMINISTIC	OPTIONAL: OPENAI
Offline, repeatable, zero API cost. Versioned scoring policy feeds transparent priority and follow-up logic.	Best-effort redacted input, strict Structured Outputs, store: false, bounded retries, and local output redaction.

Workflow

1 Bounded JSON intake Media type, byte count, object shape, source, and message length are validated.	2 Contact + classification Contacts are extracted locally; provider input and generated fields are redacted.	3 Atomic CRM write Lead, duplicate state, and event log commit in one versioned atomic state replacement.
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Built for credible review

An inspectable, production-oriented architecture with explicit privacy, security, and operating controls.

Reliable persistence	Safer interfaces
- Transactional lead + event state - OS-level interprocess file lock - Atomic tempfile + replace commit - Validated migration removes legacy PII files	- Generic server errors - HTML escaping and allowlisted classes - Formula-safe CSV cells - No-cache and browser security headers

Privacy by default

- Raw inquiry text is not retained unless STORE_RAW_MESSAGE=1 is deliberately enabled.
- API keys, provider bodies, and lead text are excluded from logs.
- Detected contact fields are best-effort redacted before OpenAI; pattern-based anonymization is not guaranteed.
- Optional bearer/session access includes TTL, logout, login throttling, CSRF, deletion, and a 90-day purge policy.
- Responses application-state storage is disabled with store: false.

Verification evidence

69 isolated tests	6 simultaneous processes	91 overall coverage (%)
Analysis, provider, lifecycle, auth, CSRF, migration, CSV, HTML, and HTTP behavior.	Mandatory regression preserves all six leads and events in one shared state.	CI gate is 90%; storage is 91%, app 87%, and provider 97%.

Client adaptation path

After requirements discovery, the same validated record can be connected to an authenticated form or inbox and written through an official CRM API. Production delivery makes authentication mandatory and adds authorization, TLS, rate limits, a managed transactional database, scheduled lifecycle automation, monitoring, backups, and reconciliation.

Integration scope

The architecture provides documented integration points for Gmail, HubSpot, Airtable, Google Sheets, Pipedrive, GoHighLevel, Zapier, Make, and n8n. Connector selection, credentials, and data mapping are configured for each client environment. No client deployment, client acceptance, live model-quality score, or measured business outcome is claimed.

Verified locally 2026-07-17 | code commit f892334ccf7aebc5eb840ea4b4675f1d55866a72 | github.com/bazhinapolly/ai-crm-lead-automation